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Graph Spatiotemporal World-Model-Driven Rolling MPC for Low-Carbon Economic Dispatch of Industrial-Park Integrated Electricity–Heat–Hydrogen Energy Systems

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Abstract

Industrial-park integrated electricity–heat–hydrogen energy systems (IEHESs) face a challenging rolling dispatch problem because strong multi-energy coupling, intertemporal storage dynamics, and forecast uncertainty make it difficult to achieve economy, low-carbon operation, and hard-constraint feasibility simultaneously. To address this issue, this paper proposes a graph spatiotemporal world-model-driven rolling model predictive control (MPC) framework, termed GraphWorldModel_MPC, for low-carbon economic dispatch of industrial-park IEHESs. First, a unified graph-based representation is constructed to characterize the topology-aware coupling relationships among the electricity, heat, and hydrogen subsystems. Second, a graph spatiotemporal world model is developed to learn multi-step state transitions, while constraint-aligned physics-consistency terms are incorporated to align the predicted trajectories with multi-energy balance, storage-boundary evolution, and ramping semantics. In addition, the learned dynamics are embedded into a hard-constrained economic MPC framework, and a quantile-based safety-tightening mechanism is adopted to mitigate residual prediction uncertainty and enhance closed-loop feasibility. Case studies on an industrial-park IEHES show that the proposed method achieves an average 24-step normalized root mean square error (NRMSE) of 4.28% and reduces the monthly total operating cost by 6.07%, 3.83%, and 10.79% compared with conventional economic MPC (EMPC), distributionally robust adaptive MPC (DRAMPC), and GRU-MPC, respectively. It also reduces equivalent carbon emissions by 6.89%, 4.52%, and 9.50% relative to these benchmarks, while maintaining zero dispatch violations in the tested monthly horizon.

Keywords: integrated electricity–heat–hydrogen energy system; graph spatiotemporal world model; model predictive control; low-carbon economic dispatch; constraint-aligned training; quantile-based safety tightening



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1. Introduction

Driven by the high penetration of renewable integration and the accelerating expansion of hydrogen utilization, integrated electricity–heat–hydrogen energy systems (IEHESs) in industrial parks increasingly require the coordinated scheduling of electricity, heat, and hydrogen under conditions of strong multi-energy coupling, pronounced intertemporal

dynamics, and substantial operational uncertainty [1]. The integration of hydrogen production, storage, and conversion further expands the flexibility space of industrial energy systems, but also introduces additional cross-sector coupling paths and state variables that must be coordinated with electricity and heat flows in real time [2,3]. In this context, dispatch is no longer a conventional single-energy cost minimization problem, but a receding-horizon decision process that must repeatedly balance economy, carbon performance, and hard-constraint feasibility [4]. Existing studies have established an important foundation for the modeling and operation of integrated energy systems [5]. Wang et al. used a coordinated scheduling framework to address the joint operation of electricity, heat, and hydrogen systems considering energy storage in heat and hydrogen pipelines, showing that explicitly modeling multi-energy flow dynamics can reveal additional operational flexibility and economic value [6]. Pan et al. used a two-stage optimization model to address integrated energy scheduling with power-to-gas and integrated demand response, showing that demand-side flexibility and multi-energy conversion can jointly improve system economy and low-carbon performance [7].

To improve adaptability under uncertainty, receding-horizon and MPC-based methods have been increasingly applied in integrated energy systems. Lv et al. [8] used model predictive control to address robust scheduling of a community integrated energy system with operational flexibility, demonstrating that rolling optimization can improve adaptability to renewable and load uncertainty while maintaining explicit constraint handling. Liu et al. [9] used a bi-level MPC-based dispatch-and-control strategy to address dynamic response performance in community integrated energy systems, showing that multi-timescale rolling optimization can better coordinate upper-level economic scheduling and lower-level real-time control. Wu et al. [10] used economic MPC to address multi-timescale operation of integrated energy systems, further showing that predictive control can exploit temporal decomposition to improve control-oriented economic performance. In the hydrogen-coupled setting, Appino et al. [11] used stochastic MPC to address real-time operation of an integrated electricity–hydrogen virtual power plant, illustrating the value of predictive control when renewable generation and market prices are uncertain. Fan et al. [12] used a distributionally robust adaptive MPC framework to address renewable uncertainty in integrated energy-system scheduling, confirming that uncertainty-aware receding-horizon formulations can enhance both robustness and economy relative to deterministic scheduling. Recent review studies on MPC in microgrids further indicate that MPC has evolved from conventional deterministic formulations toward robust, stochastic, adaptive, and learning-assisted control architectures, mainly to improve uncertainty handling and real-time operational adaptability [13]. However, these advances also reveal two persistent limitations in energy-system dispatch. First, the performance of MPC remains highly dependent on the quality of the internal predictive model. Second, robust or uncertainty-aware MPC may improve feasibility but can become conservative when the uncertainty representation is not tightly connected to the actual prediction residuals. Therefore, optimization-based rolling dispatch still requires a predictive model that can represent multi-energy dynamics accurately and a constraint-handling mechanism that can convert prediction uncertainty into executable safety margins.

In parallel with optimization-based scheduling, data-driven control has become an important research direction for integrated energy systems [14]. Recent studies have further shown that data-driven and hybrid modeling methods are increasingly used in complex integrated power and energy systems, especially when strong coupling, limited sensing, and uncertainty make purely mechanistic modeling insufficient [15,16]. These advances indicate that the latest research trend is shifting from isolated forecasting or diagnosis models toward topology-aware, uncertainty-aware, and physically interpretable learning

frameworks for complex energy-system operation. Liang et al. [17] used a Soft Actor–Critic framework to address real-time optimal scheduling of integrated energy systems for electricity, heat, and hydrogen storage, showing that the scheduling problem can be formulated as a Markov decision process and solved effectively through online policy learning. Liu et al. [18] used a deep reinforcement learning approach to address data-driven optimal scheduling of an integrated electricity–heat–gas–hydrogen energy system considering demand-side management, showing that DRL can improve adaptability in complex multi-energy environments with flexible demand-side resources. Meanwhile, graph-based spatiotemporal learning has provided a promising direction for representing networked energy systems. Su et al. [19] used a spatiotemporal convolutional graph neural network to address short-term load forecasting in regional integrated energy systems, showing that graph-based representations can better capture the coupling among multiple loads and meteorological factors than conventional sequence-only models. Yuan et al. [20] used physics-embedded graph learning to address dynamic modeling of integrated energy systems, demonstrating that topology-aware neural architectures combined with physical principles can improve the interpretability and dynamic representation capability of learned energy-system models. These studies indicate the potential of learning-based methods for representing complex energy-system dynamics. However, from a control-oriented dispatch perspective, the learned model must be recursively usable inside a constrained optimizer, where multi-step prediction errors, physical feasibility semantics, and constraint margins jointly affect closed-loop performance. Although these studies have substantially advanced integrated energy scheduling, two limitations remain insufficiently addressed in the IEHES rolling dispatch setting. First, many existing studies either rely on mechanistic predictive models [21] with limited adaptability or focus primarily on improving statistical prediction accuracy [22], but do not explicitly align the learned dynamics with the physical feasibility semantics required by downstream dispatch optimization. As a result, a trajectory that appears accurate in the mean square sense may still deviate from the electricity/heat/hydrogen balance relationships, storage-boundary evolution, and ramping logic that define the true feasible operating region. Second, even when a learned predictive model is coupled with rolling optimization, most existing approaches still optimize under nominal state and power bounds without explicitly transforming prediction-error statistics into executable safety margins [23–25]. In tightly coupled electro–heat–hydrogen dispatch, this omission is particularly critical because accumulated prediction residuals can propagate across storage states, conversion pathways, and network balances, thereby causing nominally feasible decisions to become infeasible during real closed-loop operation.

To further clarify the distinction between the proposed work and recent studies, Table 1 summarizes the representative literature published in recent years from the perspectives of method, uncertainty handling, system type, objective, data source, and main limitation.

As shown in Table 1, recent studies have made important progress in robust MPC, multi-timescale optimization, data-driven MPC, graph-based forecasting, and physics-embedded energy-system modeling. However, most existing methods either focus on optimization with predefined system dynamics, uncertainty-aware scheduling without topology-aware learned transitions, or graph-based prediction without direct integration into a hard-constrained economic MPC. In contrast, the proposed GraphWorldModel_MPC explicitly embeds the learned graph spatiotemporal transition model into the MPC as a transition constraint and further converts empirical prediction residuals into quantile-tightened state and power bounds. Therefore, the main difference in this work lies in the MPC-level integration of learned multi-energy transition dynamics, constraint-aligned rollout training, and residual-aware feasible-region tightening for industrial-park IEHES dispatch.

Table 1. Comparative summary of recent studies on MPC and learning-based dispatch for integrated energy systems.

Ref.	Method	System Type	Uncertainty Handling	Objective	Data Source/Case	Main Limitation Relative to This Work
Wang et al. [6]	Coordinated scheduling with heat and hydrogen pipeline storage	Integrated electricity–heat–hydrogen system	Mainly deterministic scheduling	Economic coordination and storage flexibility	Numerical IES case	Does not embed learned transition dynamics into MPC or address prediction-error-driven safety tightening
Pan et al. [7]	Two-stage optimization with P2G and integrated DR	Integrated energy system with P2G	Scenario-based or deterministic uncertainty treatment	Multi-objective economic and low-carbon scheduling	Simulation case	Focuses on planning/operation optimization, but not on graph-based multi-step transition learning for rolling MPC
Fan et al. [12]	Distributionally robust adaptive MPC	Integrated energy system	Distributionally robust uncertainty modeling for renewable uncertainty	Robust economic scheduling	IES benchmark case	Improves robustness, but does not learn topology-aware electricity–heat–hydrogen transition dynamics
Dong et al. [24]	Robust real-time scheduling with multi-step interval prediction	Integrated energy system	Interval prediction of uncertain sources and loads	Real-time robust energy scheduling	Simulation-based IES case	Uses interval prediction for uncertainty, but does not formulate a graph world model as an explicit MPC transition constraint
Wei et al. [25]	Two-layer multi-timescale rolling optimization	Electric–hydrogen hybrid energy storage system	Renewable uncertainty considered in rolling optimization	Multi-timescale electric–hydrogen scheduling	Simulation case	Focuses on electric–hydrogen storage coordination and does not consider full electricity–heat–hydrogen graph transition learning
Chen et al. [26]	Data-driven learning-based MPC	Energy-intensive system	Data-driven prediction within MPC	Energy-saving control	Industrial energy-intensive process data	Learning-based MPC is considered, but without multi-energy graph representation, physics-consistent rollout training, or quantile tightening
Su et al. [19]	Spatiotemporal graph neural network	Regional integrated energy system	Forecasting-oriented treatment	Short-term multi-energy load forecasting	Regional IES data	Focuses on prediction accuracy, not on embedding learned dynamics into constrained economic MPC
Yuan et al. [20]	Physics-embedded graph learning	Integrated energy system	Mainly modeling-oriented	Energy-system dynamic modeling	Simulation and modeling data	Improves graph-based physical modeling, but does not develop a hard-constrained rolling MPC with residual-based safety margins
This work	Graph spatiotemporal world-model-driven rolling MPC	Industrial-park electricity–heat–hydrogen IEHES	Multi-source forecast uncertainty and quantile-based safety tightening	Low-carbon economic dispatch with feasibility preservation	Hourly industrial-park IEHES time-series data	Embeds topology-aware learned transition dynamics as MPC constraints and converts prediction residuals into variable-specific safety margins

Based on the above review, the research gap addressed in this paper can be summarized as follows: existing rolling dispatch studies for integrated energy systems still lack a unified framework that can simultaneously learn topology-aware multi-energy dynamics,

preserve the physical feasibility semantics of multi-step predicted trajectories, and explicitly account for residual prediction uncertainty in hard-constrained economic MPC. This gap is particularly important for industrial-park IEHESs, where electricity–heat–hydrogen coupling, storage carry-over dynamics, and renewable/load forecast errors jointly affect the feasibility and economy of online dispatch. Existing graph-based or learning-based energy-system studies have mainly focused on forecasting accuracy, representation learning, or policy learning, whereas the present work focuses on the interface between learned multi-step transition prediction and hard-constrained economic MPC. In this setting, the learned model must not only predict future states accurately, but also produce rollout trajectories that remain usable inside a constrained optimizer. Therefore, the distinction of this work is not the isolated use of graph learning, MPC, or physics-consistency terms, but their integration into a rolling dispatch pipeline in which topology-aware prediction, constraint-aligned rollout training, and quantile-based constraint tightening are jointly evaluated under the same MPC interface.

More specifically, the MPC-related novelty of this work lies in the formulation of a learned-dynamics-constrained economic MPC for industrial-park IEHES dispatch. Instead of using the learned model only as an external forecaster, the graph spatiotemporal world model is embedded into the MPC as an explicit transition constraint, so that each candidate control sequence is evaluated through recursively generated electricity–heat–hydrogen state trajectories. In addition, the prediction residuals of the learned transition model are converted into quantile-based tightened state and power bounds, which makes the feasible region of the MPC directly aware of model uncertainty. Therefore, the proposed method differs from a simple combination of graph learning and MPC; the learned dynamics determine the admissible predicted trajectories, while the error-aware tightening determines the executable constraint margins used by the optimizer. This provides a concrete MPC-level distinction from conventional nominal-model MPC, robust adaptive MPC, and sequence-only learning-based MPC.

The objective of this paper is therefore to develop a graph spatiotemporal world-model-driven rolling MPC framework for low-carbon economic dispatch of industrial-park IEHESs, with the aim of improving multi-step predictive fidelity, closed-loop feasibility, and economic-carbon performance under forecast uncertainty. Accordingly, the central research question of this paper is: can a topology-aware learned transition model, when embedded as an explicit transition constraint in a hard-constrained economic MPC, improve multi-step prediction accuracy, closed-loop feasibility, and economic-carbon dispatch performance compared with conventional nominal-model MPC, robust adaptive MPC, and sequence-only learning-based MPC? This question is tested through three measurable aspects: 24-step state-transition prediction accuracy, monthly operating cost and carbon emissions, and dispatch feasibility under nominal and perturbed operating conditions. Therefore, the contribution of this work is not limited to proposing another forecasting model or another MPC variant, but lies in verifying whether learned graph-based multi-energy dynamics can be effectively translated into executable MPC constraints for industrial-park IEHES dispatch. The specific contributions are summarized as follows.

1. A graph-based IEHES representation is constructed to encode the directed electricity–heat–hydrogen coupling topology. Based on this representation, a graph spatiotemporal world model is developed to learn multi-step state transitions by combining topology-aware message passing and temporal rollout modeling.
2. A constraint-aligned training strategy is introduced for the learned world model. In addition to conventional multi-step prediction loss, balance-, storage-boundary-, and ramping-consistency terms are incorporated to make the predicted trajectories more compatible with the feasibility requirements of downstream rolling optimization.

3. A learned-dynamics-constrained economic MPC formulation is developed, in which the graph spatiotemporal world model is embedded as an explicit transition constraint rather than used only as an external predictor. Under this formulation, the optimizer searches the control sequence subject to recursively learned electricity–heat–hydrogen state evolution, multi-energy balance constraints, device limits, ramping constraints, and DR contractual constraints. In addition, empirical quantiles of rolling prediction residuals are transformed into tightened state and power bounds, making the MPC feasible region directly aware of model uncertainty.

The remainder of this paper is organized as follows. Section 2 formulates the IEHES dispatch problem, including the system model, state/control/exogenous variables, operating constraints, and the finite-horizon economic dispatch objective. Section 3 presents the graph spatiotemporal world model and its training objective. Section 4 introduces the world-model-driven rolling MPC framework and the quantile-based safety-tightening mechanism. Section 5 reports comparative, robustness, ablation, and sensitivity results based on an industrial-park case in Inner Mongolia. Finally, Section 6 discusses the limitations of the proposed framework, and Section 7 concludes the paper and outlines future research directions.

2. System Model and Problem Formulation

This section formulates the industrial-park IEHES dispatch problem from three levels. First, the physical boundary and electricity–heat–hydrogen coupling structure are defined to specify the controllable devices, storage units, loads, and external inputs considered in the study. Second, the operational information is organized into state, control, and exogenous variables, which provides a unified interface for the graph spatiotemporal world model and the downstream MPC. Third, the multi-energy balance equations, device operating constraints, and finite-horizon economic objective are established to define the feasible rolling dispatch problem. This formulation links the physical IEHES structure with the learned transition model and the hard-constrained MPC developed in the following sections.

2.1. System Boundary and Multi-Energy Coupling Structure

The studied system is an industrial-park integrated electricity–heat–hydrogen energy system (IEHES), whose physical boundary is shown in Figure 1. The system includes the utility grid interface, photovoltaic (PV) and wind generation units, battery energy storage with state of charge (SOC), thermal energy storage (TES), hydrogen storage tanks, industrial electric loads, thermal loads, hydrogen demand, and demand response (DR) resources. Unlike conventional single-energy systems, the electricity, heat, and hydrogen subsystems are not operated independently. Instead, they are tightly coupled through energy conversion devices and storage units. In particular, the CHP unit acts as the main electricity–heat coupling device by converting natural gas into both electric and thermal outputs, whereas the electrolyzer serves as the key electricity–hydrogen coupling device by converting electricity into hydrogen. Meanwhile, battery energy storage, thermal energy storage, and hydrogen storage provide intertemporal flexibility on the electricity side, heat side, and hydrogen side, respectively. External market and environmental information, including electricity price, gas price, carbon price, renewable forecasts, load forecasts, and hydrogen demand forecasts, jointly constitute the exogenous inputs of the rolling dispatch problem.

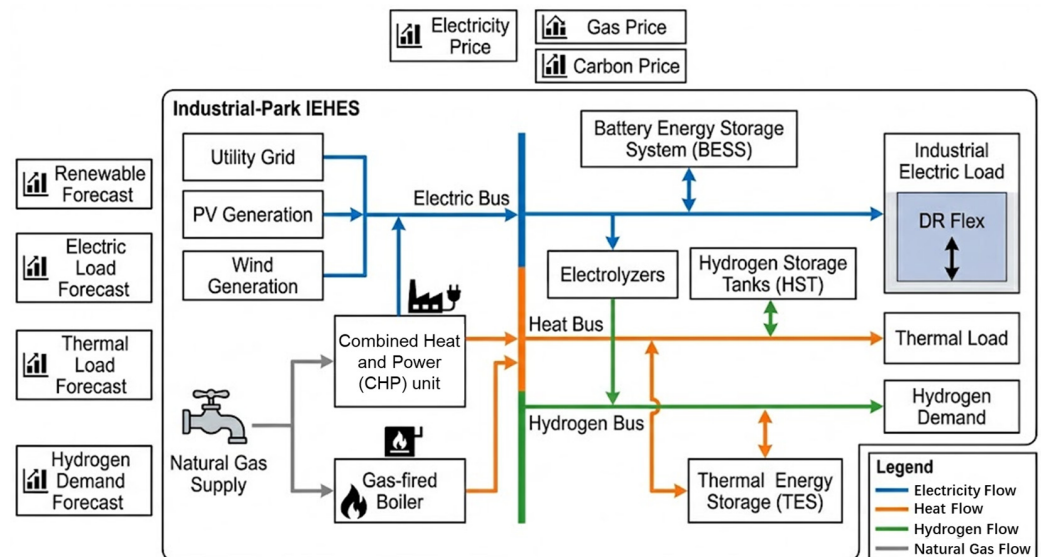


Figure 1. Industrial-park IEHES architecture with electricity–heat–hydrogen coupling among grid, renewables, CHP, boiler, electrolyzer, storage units, loads, and DR resources.

To provide a unified representation for subsequent dynamic learning and optimization, the industrial-park IEHES is abstracted as a directed graph $G = (V, E)$, where the node set V represents devices, storage units, energy hubs, or aggregated loads, and the edge set E describes energy-transfer or coupling relationships. According to the carrier type, the edge set can be further divided into the electricity edge set E^{el} , the heat edge set E^{th} , and the hydrogen edge set E^{h2} . This graph abstraction preserves the local topology, coupling paths, and message-passing structure of the multi-energy network, establishing a consistent foundation for the subsequent graph-based world model. Accordingly, over the discrete time horizon $t = 1, \dots, T$, the true system evolution can be written as:

$$s_{t+1} = f(s_t, u_t, w_t) \quad (1)$$

where s_t denotes the system state, u_t denotes the control input, w_t denotes the exogenous input, and $f(\cdot)$ denotes the underlying state-transition dynamics of the industrial-park IEHES.

Based on the above physical boundary and graph abstraction, the dispatch problem is expressed in a compact state-space form. This state-space description is not intended to replace the physical constraints of the IEHES; rather, it provides a unified input–output interface through which the subsequent world model can learn system transitions and the MPC can evaluate candidate control sequences. To maintain this unified interface, the operational information of the IEHES is divided into three categories: state variables, control variables, and exogenous variables. The state vector s_t is used to describe the operating condition of the system at time t . It mainly consists of electricity-side power variables and storage states, heat-side supply and thermal-storage states, hydrogen-side flow and storage states, together with market and environmental quantities that have already entered the current state. In a compact form, it can be written as:

$$s_t = [P_t^{el}, SOC_t, Q_t^{th}, H_t^{th}, \dot{m}_t^{h2}, M_t^{h2}, \lambda_t^e, \lambda_t^g, \lambda_t^c]^T \quad (2)$$

where P_t^{el} denotes electricity-related state variables; SOC_t denotes the state of charge of battery energy storage; Q_t^{th} and H_t^{th} denote heat-flow and thermal-storage states; $\dot{m}_t^{h2}$ and M_t^{h2} denote hydrogen flow and hydrogen inventory states; and λ_t^e , λ_t^g , and λ_t^c denote electricity price, gas price, and carbon-related cost signals, respectively. These variables are selected because they determine the current operating condition, storage carry-over status,

and economic environment of the IEHES, and therefore provide the necessary information for both transition prediction and rolling optimization.

The control vector u_t represents the dispatch decisions directly applied by the scheduler at time t , including the charging/discharging power of battery storage, the electric and thermal outputs of the CHP, the thermal output of the gas-fired boiler, the power consumption of the electrolyzer, the charging/discharging actions of thermal and hydrogen storage, and the DR activation level. A generic form is given by:

$$u_t = [P_t^{ch}, P_t^{dis}, P_t^{CHP,e}, Q_t^{CHP,h}, Q_t^{GB}, P_t^{el}, q_t^{th}, q_t^{h2}, u_t^{DR}]^T \quad (3)$$

These variables constitute the direct decision space of the scheduler and determine how energy is converted, stored, supplied, or shifted at each dispatch interval.

The exogenous vector w_t contains quantities that are not directly controllable but affect the future system evolution, including renewable generation, electricity/heat/hydrogen demand, and market-related signals over the prediction horizon. It is expressed as:

$$w_t = [\hat{P}_t^{RES}, \hat{L}_t^{el}, \hat{L}_t^{th}, \hat{L}_t^{h2}, \hat{\lambda}_t^e, \hat{\lambda}_t^g, \hat{\lambda}_t^c]^T \quad (4)$$

where $\hat{P}_t^{RES}$, $\hat{L}_t^{el}$, $\hat{L}_t^{th}$, and $\hat{L}_t^{h2}$ denote the forecast renewable output, electric load, thermal load, and hydrogen demand, respectively, while $\hat{\lambda}_t^e$, $\hat{\lambda}_t^g$, and $\hat{\lambda}_t^c$ denote the forecast electricity price, gas price, and carbon-related signals within the prediction horizon. These quantities are treated as forecast inputs in the MPC horizon because they affect future system evolution but cannot be directly manipulated by the controller. Therefore, the feasible control set at state s_t can be defined as:

$$\mathcal{A}(s_t) = \{u_t \in \mathcal{U} | (s_t, u_t) \in \mathcal{Z}_t\} \quad (5)$$

where $\mathcal{U}$ denotes the control space and $\mathcal{Z}_t$ denotes the feasible set satisfying all physical and operational constraints at time t .

To maintain consistency with the graph-based dynamic model, the local feature vector of node t is further defined as

$$x_{i,t} = [p_{i,t}, SOC_{i,t}, h_{i,t}, \lambda_t, \hat{d}_{i,t}, \hat{r}_{i,t}] \quad (6)$$

where $p_{i,t}$, $SOC_{i,t}$, and $h_{i,t}$ denote the local power, storage-related, and heat-related features, respectively; λ_t denotes market-dependent exogenous information; and $\hat{d}_{i,t}$, $\hat{r}_{i,t}$ denote demand-side and renewable-side predictive inputs. For nodes that do not possess certain attributes, zero-padding or masking is adopted to preserve a unified feature dimension. This representation enables the subsequent world model to simultaneously exploit temporal information and the topology-aware multi-energy coupling structure of the IEHES.

2.2. Operational Constraints and Economic Objective

The feasible rolling dispatch problem is determined by three groups of constraints: multi-energy balance constraints, device-level operating constraints, and contractual DR constraints. The balance constraints ensure carrier-level conservation of electricity, heat, and hydrogen at each time step. Device-level constraints describe storage dynamics, charging/discharging exclusiveness, output limits, and ramping limits. DR constraints restrict the amount and timing of demand adjustment according to the predefined contractual rules. First, the electricity, heat, and hydrogen carriers must each satisfy their corresponding balance equations. For compactness, they are written in the following unified form:

$$\phi^{el}(s_t, u_t, w_t) = 0, \phi^{th}(s_t, u_t, w_t) = 0, \phi^{h2}(s_t, u_t, w_t) = 0 \quad (7)$$

where ϕ^{el} , ϕ^{th} , and ϕ^{h2} denote the electricity, heat, and hydrogen balance residuals at time t , respectively. These balance equations reflect the essential feature of the problem: a decision made in one energy carrier may propagate to the others through coupling devices such as the CHP, electrolyzer, and storage units.

Second, the system is subject to device power limits, storage dynamics, mutual exclusiveness of charging and discharging, ramping limits, and DR contractual rules. Taking battery energy storage as an example, its state evolution can be expressed as

$$SOC_{t+1} = SOC_t + \frac{\eta^{ch} P_t^{ch} \Delta t}{E^{bat}} - \frac{P_t^{dis} \Delta t}{\eta^{dis} E^{bat}} \quad (8)$$

where E^{bat} is the rated battery capacity, and η^{ch} , η^{dis} are the charging and discharging efficiencies, respectively. Thermal energy storage and hydrogen storage can be modeled in the same manner. The corresponding box constraints on states and controls are written as:

$$\underline{s} \leq s_t \leq \bar{s}, \underline{u} \leq u_t \leq \bar{u} \quad (9)$$

and the ramping constraint is written as

$$|u_t - u_{t-1}| \leq \bar{r} \Delta t \quad (10)$$

where $\bar{r}$ denotes the ramping limit of the controllable devices. For CHP units and other coupled conversion devices, additional feasible-region constraints must be imposed to ensure the admissible joint operation of multiple outputs. For the DR module, contractual constraints such as dispatchable upper bounds, energy windows, and settlement-related rules must also be satisfied. Accordingly, the feasible set $\mathcal{Z}_t$ is jointly determined by the balance equations, storage dynamics, device bounds, coupling feasibility regions, and DR rules.

In this formulation, a candidate control input is feasible only if the resulting system state satisfies all carrier-balance equations, device operating limits, storage-state bounds, ramping constraints, and DR contractual constraints at the same dispatch interval. For the rolling-horizon problem, feasibility further requires that these constraints remain satisfied for all predicted steps over the MPC horizon. Therefore, the feasible region is not defined by independent box constraints alone, but by the joint interaction among multi-energy balances, storage dynamics, device limits, and coupling relations.

From the economic perspective, the rolling dispatch objective is to minimize the total operating cost over the prediction horizon. The single-step operating cost is composed of grid electricity cost, gas consumption cost, carbon cost, contractual penalty cost, and DR settlement cost, namely:

$$C_t = C_t^{grid} + C_t^{gas} + C_t^{carbon} + C_t^{penalty} + C_t^{dr} \quad (11)$$

Given a prediction horizon of length N_p , the finite-horizon economic objective at time t is defined as

$$\min \sum_{\tau=t}^{t+N_p-1} C_\tau \quad (12)$$

The optimization problem is subject to Equations (7)–(10) and all related device and contractual constraints. It should be emphasized that the economic objective in this paper is not a simple electricity cost minimization problem. Instead, it is a multi-term joint optimization problem simultaneously influenced by gas consumption, carbon accounting, DR settlement, and contractual penalties. This motivates the need for a dispatch frame-

work that can simultaneously ensure predictive fidelity, physical feasibility, and economic optimality. The concrete numerical settings adopted in the dispatch model are summarized in Table 2.

Table 2. Equipment and constraint parameters.

Item	Value	Item	Value
Utility grid import limit	10 MW	Battery SOC bounds	0.1–0.9
PV installed capacity	2.5 MW	Thermal energy storage capacity	4 MW _{th}
Wind installed capacity	3.0 MW	TES power limit	1.5 MW _{th}
CHP electric output	0.8–4.0 MW	TES efficiency	0.92
CHP thermal output	1.2–4.5 MW _{th}	TES state bounds	0.1–0.9
CHP electric ramping limit	1 MW/h	Hydrogen storage capacity	1.5 t
CHP thermal ramping limit	1.2 MW _{th} /h	Hydrogen flow limit	0.08 t/h
Gas-fired boiler output	6 MW _{th}	DR shift capacity	0.13 MW
Gas-fired boiler ramping limit	1.5 MW _{th} /h	DR curtailment capacity	0.07 MW
Gas-fired boiler efficiency	0.90	Electricity purchase price	0.42/0.54/0.72/0.88 RMB/kWh
Electrolyzer power	2 MW	Gas price	3.5 RMB/Nm ³
Electrolyzer ramping limit	0.5 MW/h	Carbon price	80 RMB/tCO ₂
Electrolyzer hydrogen conversion coefficient	20 kg/MWh	Penalty coefficient	0.20 RMB/kWh
Battery energy capacity	4 MWh	MPC prediction horizon	24
Battery power limit	1.0 MW		

3. Graph Spatiotemporal World Model

To provide an intuitive overview of the proposed predictive architecture before presenting the mathematical details, Figure 2 illustrates the overall structure of the graph spatiotemporal world model used in this study. Starting from historical state, control, and exogenous-input trajectories, the model first constructs topology-aware node representations based on the directed IEHES graph, then extracts spatial coupling and temporal dependency through graph encoding and sequence modeling, and finally performs one-step state decoding and autoregressive multi-step rollout. In addition to the conventional prediction loss, physics-consistency terms are incorporated into training so that the learned trajectories remain better aligned with electricity–heat–hydrogen balance, storage-boundary evolution, and ramping semantics required by downstream dispatch optimization.

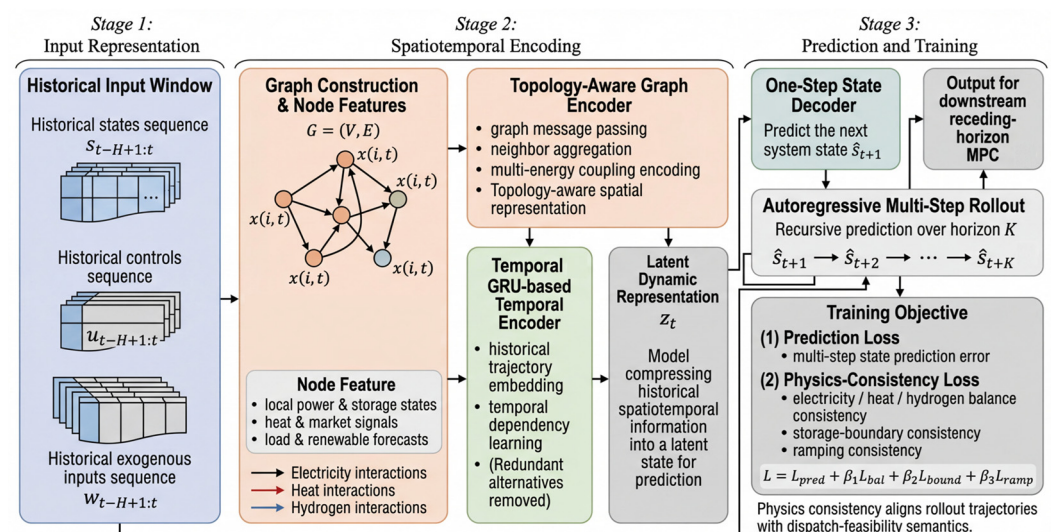


Figure 2. Graph spatiotemporal world model for multi-step IEHES state prediction using graph encoding, temporal encoding, and recursive rollout.

3.1. Problem Setting and Input–Output Formulation

The true transition dynamics $f(\cdot)$ in Equation (1) are generally unknown or only partially captured by simplified mechanistic models. In industrial-park IEHESs, the system evolution is jointly influenced by multi-energy coupling, storage carry-over effects, demand response adjustments, and time-varying exogenous inputs. Under such conditions, a purely mechanistic model may be insufficiently adaptive, whereas a purely statistical predictor may fail to preserve the operational semantics required by downstream dispatch optimization. To address this issue, a parameterized graph spatiotemporal world model f_θ is constructed to approximate the underlying transition process from historical operational data. Rather than directly generating dispatch decisions, the model is trained to predict future system states and then serves as the internal predictive kernel of the rolling MPC described in Section 4. This design is consistent with the core problem setting of this paper, in which the dispatch task is solved through learned transition modeling and hard-constrained rolling optimization rather than end-to-end policy learning.

Let the historical window length be H . At time t , the input to the world model consists of the recent state trajectory, control trajectory, and exogenous trajectory over the interval $[t - H + 1, t]$. The one-step prediction is written as

$$\hat{s}_{t+1} = f_\theta(s_{t-H+1:t}, u_{t-H+1:t}, w_{t-H+1:t}) \quad (13)$$

where $\hat{s}_{t+1}$ denotes the predicted next state. Compared with a one-shot vector predictor, this formulation explicitly exploits the temporal dependence created by storage dynamics, ramping constraints, and inter-energy coupling, and therefore is better suited to the receding-horizon nature of the IEHES dispatch problem. Moreover, because the system has already been represented as a graph $G = (V, E)$ with electricity, heat, and hydrogen carrier edges, the state trajectory entering the world model is not treated as an unordered sequence. Instead, the local states of individual devices and hubs are mapped to node-level features and then processed by graph-based spatial encoders before temporal aggregation. This allows the predictor to preserve the physical topology of the IEHES rather than flattening it into a purely sequence-based approximation.

3.2. Topology-Aware Graph Representation

Given the directed graph $G = (V, E)$, each node $i \in V$ is associated with a feature vector $x_{i,t}$ at time t as defined in Equation (6). The node features include local power-related quantities, storage states, heat-related quantities, market-dependent exogenous information, and demand/renewable prediction inputs. Since different device types do not share identical physical attributes, missing entries are handled through masking or zero-padding so that all nodes can be embedded into a common latent space. In this way, the graph representation remains compatible with heterogeneous system components while preserving a unified tensor structure for training. The purpose of this representation is not merely to reduce dimensionality, but to encode the fact that the operating state of each device is influenced by its neighbors through directed energy-transfer paths and physical coupling relations. In the industrial-park IEHES, such topology-dependent interactions are fundamental. For example, a change in CHP dispatch simultaneously affects the electricity and heat subsystems, whereas electrolyzer operation links electricity consumption to hydrogen production and storage evolution.

To extract topology-aware embeddings, the graph encoder performs L layers of directed attention-weighted message passing at each time step t . First, the input feature of node i is projected into the latent space to obtain the initial hidden representation. Then, at the l -th graph layer, each incoming neighbor $j \in \mathcal{N}(i)$ sends a message to node i through a learnable linear transformation of its current hidden feature. To distinguish the unequal in-

fluence of different neighbors and coupling paths, an attention score is further computed for each directed edge (j, i) based on the current hidden features of both the source node and the target node, together with the corresponding edge attribute. Here, the edge attribute is used to distinguish electricity, heat, hydrogen, or cross-carrier coupling relations, so that the graph encoder can preserve the physical meaning of heterogeneous interconnections rather than treating all edges identically. The resulting attention scores are normalized over all incoming neighbors of node i , thereby yielding attention coefficients that quantify the relative contribution of each incoming message.

Accordingly, the hidden representation of node i at the next layer is updated by combining two parts: the transformed self-feature of node i and the weighted aggregation of all incoming neighbor messages. The self-feature term preserves the intrinsic operating condition of the target node, while the aggregated message term captures the influence propagated from structurally connected components through directed energy-transfer paths. In this way, the message-passing mechanism explicitly reflects the physical topology of the IEHES. Different coupling paths can contribute unequally to the target-node embedding, and this is particularly important in the present problem because the operational dependence among components is not spatially uniform. For instance, CHP-related electricity–heat coupling, electrolyzer-related electricity–hydrogen coupling, and storage-related intertemporal support all have different structural roles in the system and should therefore be weighted differently during representation learning.

Formally, denoting the hidden representation of node i at the l -th layer and time step τ by $h_{i,\tau}^{(l)}$, the update can be written as

$$h_{i,\tau}^{(l+1)} = \sigma(W_1 h_{i,\tau}^{(l)} + \sum_{j \in \mathcal{N}(i)} a_{ij,\tau} W_2 h_{j,\tau}^{(l)}) \quad (14)$$

where $\mathcal{N}(i)$ denotes the neighbor set of node i , W_1 and W_2 are trainable parameters, $a_{ij,\tau}$ denotes a normalized edge weight or attention coefficient, and $\sigma(\cdot)$ is a nonlinear activation function. Through repeated aggregation over neighbors, the graph encoder enables each node representation to absorb both local state information and the influence of structurally related components. This mechanism is particularly useful in multi-energy systems because operational coupling is not uniform across all state dimensions, but is concentrated along specific physical pathways. The output of the graph encoder at time τ is denoted by $H_\tau = \{h_{i,\tau}^{(L)}\}_{i \in V}$, which summarizes the topology-aware system representation at that time step.

3.3. Temporal Encoding and Multi-Step Rollout

Although graph message passing captures the spatial coupling structure at each time step, it does not itself account for the temporal dependencies induced by storage carry-over, device ramping, and time-varying exogenous inputs. Therefore, the sequence of graph representations over the historical window is further processed by a temporal encoder. Let the graph-encoded sequence over $[t - H + 1, t]$ be $\{H_{t-H+1}, \dots, H_t\}$. A temporal encoder, implemented as either a gated recurrent unit (GRU) or a Transformer-style sequence model, maps this sequence into a compact latent representation z_t namely:

$$z_t = \text{TemporaEncoder}(H_{t-H+1}, \dots, H_t) \quad (15)$$

This latent representation summarizes the recent operating trajectory of the industrial-park IEHES and serves as the basis for state decoding. The next state is then obtained by a linear decoder

$$\hat{s}_{t+1} = W_d z_t + b_d \quad (16)$$

where W_d and b_d are trainable parameters. The role of the temporal encoder is not only to improve one-step accuracy, but also to preserve intertemporal information required for stable multi-step rollout. In the present IEHES dispatch problem, temporal encoding is necessary because the same instantaneous graph state may correspond to different future evolutions depending on previous storage charging/discharging directions, ramping history, and lagged exogenous variations. For example, two operating points with similar electricity–heat–hydrogen balances can lead to different subsequent SOC, TES, or hydrogen-inventory trajectories if their preceding trajectories are different. Therefore, the temporal encoder acts as a dynamic memory module that transforms the graph-encoded historical trajectory into a latent state representation for transition prediction. Without this temporal module, the predictor degenerates into a snapshot-based graph model and tends to lose the carry-over effects of storage dynamics and device ramping, which may lead to accumulated errors during recursive rollout. Since the downstream MPC relies on future state trajectories rather than isolated one-step predictions, the predictor must maintain meaningful structure under recursive application.

Accordingly, during online rolling dispatch, the learned world model is used in an autoregressive fashion over the prediction horizon N_p . Starting from the current observed state $\hat{s}_t = s_t$, the future state trajectory is generated as:

$$\hat{s}_{t+k+1} = f_{\theta}(\hat{s}_{t+k}, u_{t+k}, \bar{w}_{t+k}), \quad k = 0, \dots, N_p - 1 \quad (17)$$

where $\bar{w}_{t+k}$ denotes the forecast exogenous input at future step $t + k$.

It should be noted that the multi-step rollout is used not only during online MPC prediction, but also during world model training. During training, the predicted state at one rollout step is recursively fed into the next prediction step together with the recorded control and exogenous inputs, and the resulting multi-step trajectory is supervised by the corresponding ground-truth trajectory. This training strategy reduces the mismatch between one-step supervised learning and the recursive use of the model in MPC. In other words, the model is explicitly trained under the same rollout mechanism as that used by the downstream rolling optimizer, which improves long-horizon stability and mitigates error accumulation.

This recursive rollout is the key interface through which the world model is connected to the MPC layer. In other words, the world model does not output a final dispatch decision. Instead, it supplies a learned predictive state-transition operator that allows the economic optimizer to evaluate candidate control sequences under hard constraints.

3.4. Physics-Consistent Training Objective

A world model trained only by minimizing statistical prediction error may generate trajectories that are numerically close to historical data while still violating the physical semantics required by dispatch optimization. For example, a predictor may achieve low mean square error but still produce state trajectories that are inconsistent with electricity/heat/hydrogen balance, storage-boundary evolution, or device ramping behavior. To mitigate this mismatch, the training objective combines a multi-step prediction loss with a set of physics-consistency penalties that are deliberately aligned with the operational constraints introduced in Section 2. In this paper, the physics-consistency terms are used as a constraint-aligned training component rather than as an independent methodological novelty. Their purpose is to make the learned rollout trajectories more compatible with the feasibility semantics required by the downstream hard-constrained MPC, especially when the model is recursively applied over a multi-step prediction horizon. Therefore, this paper does not claim physics-consistency regularization as a new theoretical contribution. Instead, its role is empirically evaluated through the ablation study in Section 5.6, where

the same graph architecture and MPC formulation are compared with and without this constraint-aligned training component.

Let the training rollout length be K . The multi-step prediction loss is defined as:

$$L_{pred} = \sum_{k=1}^K a_k \|\hat{s}_{t+k} - s_{t+k}\|_2^2 \quad (18)$$

where a_k is the weight assigned to the k -step prediction error. To regularize the rollout trajectory with respect to the system's physical feasibility semantics, the physics-consistency term is written as:

$$L_{phys} = \beta_1 L_{bal} + \beta_2 L_{bound} + \beta_3 L_{ramp} \quad (19)$$

where L_{bal} penalizes electricity/heat/hydrogen balance residuals, L_{bound} penalizes storage-boundary deviations, and L_{ramp} penalizes ramping inconsistency. The overall training objective is then:

$$L = L_{pred} + L_{phys} \quad (20)$$

The three physics-consistency terms play complementary roles. L_{bal} encourages the rollout trajectory to remain close to the conservation structure of multi-energy flows, L_{bound} discourages boundary-inconsistent storage evolution, and L_{ramp} suppresses unrealistic step-to-step transitions of controllable variables. Together, these terms make the learned world model better aligned with the feasible operating manifold of the IEHES and therefore more suitable as a predictive kernel for downstream rolling optimization. The hyperparameters of the world model were selected using the validation segment described in Section 5.1. The hidden dimension, number of graph layers, number of GRU layers, dropout rate, learning rate, batch size, and loss weights were tuned within a limited candidate range, and the final setting was selected according to the validation 24-step prediction error and rollout stability. During training, the validation loss was monitored after each epoch, and the model checkpoint with the lowest validation rollout error was retained for subsequent MPC evaluation. Gradient clipping and weight decay were used to reduce unstable parameter updates and overfitting. The training process was regarded as converged when the validation rollout loss no longer showed a visible decreasing trend over consecutive epochs.

For computational complexity, the graph encoder mainly scales with the number of directed edges and graph layers, while the temporal encoder scales with the historical window length and hidden dimension. Since the studied industrial-park IEHES has a moderate number of devices and coupling edges, the trained world model can generate 24-step rollout trajectories efficiently for the downstream MPC. To improve implementation-level clarity, the main variables, constraints, graph components, and loss terms used in the proposed framework are further specified as follows. The state vector contains dispatch-related power variables and storage states, including grid import, CHP electric and thermal outputs, gas-boiler output, electrolyzer power, battery SOC, TES state, hydrogen inventory, and storage charging/discharging quantities. The control vector contains the directly optimized dispatch actions, including battery charging/discharging power, CHP output commands, gas-boiler thermal output, electrolyzer power, TES and hydrogen-storage charging/discharging actions, and DR activation. The exogenous vector contains forecast PV output, wind output, electric load, thermal load, hydrogen demand, electricity price, gas price, and carbon-related price signal. In the graph implementation, each physical device, storage unit, energy hub, or aggregated load is treated as a node, and each electricity, heat, hydrogen, or cross-carrier coupling path is treated as a directed edge with a carrier-type attribute. Missing node attributes are handled by zero-padding and masking. During training, the multi-step prediction loss is computed over all selected state variables over the 24-step rollout horizon. The balance-consistency loss is calculated from electricity-,

heat-, and hydrogen-balance residuals, the boundary-consistency loss penalizes predicted storage states outside admissible SOC/TES/hydrogen-inventory ranges, and the ramping-consistency loss penalizes unrealistic step-to-step variations in controllable outputs. During MPC implementation, the equality constraints include learned transition dynamics and multi-energy balance constraints, while the inequality constraints include device output bounds, storage bounds, ramping limits, DR contractual limits, and quantile-tightened bounds for selected critical variables.

For readability and reproducibility, the main architectural choices and training hyperparameters of the graph spatiotemporal world model are summarized in Table 3.

Table 3. The architecture and training hyperparameters of the graph spatiotemporal world model.

Item	Value	Item	Value
Historical window length H	24	GRU hidden dimension	128
Prediction horizon in training K	24	Dropout rate	0.10
Prediction horizon in MPC N_p	24	Prediction loss	Multi-step MSE
Node feature dimension d_{in}	6	Physics-consistency terms	Balance + storage-boundary + ramping
Graph type	Directed graph	Loss weights $\beta_1 \beta_2 \beta_3$	1.0 0.5 0.5
Number of graph layers L	2	Optimizer	Adam
Hidden dimension d_h	128	Learning rate	1×10^{-3}
Aggregation type	Attention-weighted message passing	Batch size	256
Activation function	ReLU	Number of epochs	200
Encoder type	GRU	Gradient clipping	1.0
Number of GRU layers	2	Weight decay	1×10^{-5}

4. World-Model-Driven Rolling MPC and Quantile-Based Safety Tightening

To clarify how the learned transition model is integrated into online dispatch, Figure 3 presents the overall world-model-driven rolling MPC framework adopted in this study. At each control step, the current measured state and forecast exogenous sequence are used to generate future state trajectories through the graph spatiotemporal world model, while empirical quantiles of rolling prediction errors are used to tighten critical state and power bounds. The tightened trajectories and constraints are then embedded into a hard-constrained economic MPC, whose first optimal control input is implemented and fed back to the physical system in a receding-horizon manner.

4.1. Interface Between the Learned World Model and Rolling Dispatch

After the graph spatiotemporal world model f_θ is trained as described in Section 3, it is embedded into a receding-horizon model predictive control framework to support online dispatch of the industrial-park IEHES. The learned model provides a data-driven approximation of the system transition, while the MPC layer computes economically optimal control actions under explicit hard constraints. This separation allows the proposed framework to combine adaptive dynamic prediction with physically feasible rolling optimization.

At each control instant t , the current measured system state s_t is treated as known, and the future exogenous sequence over the prediction horizon is provided by the forecast vector $\tilde{w}_{t:t+N_p-1}$. The learned world model is then used to recursively propagate the future state trajectory under a candidate control sequence. Let the prediction horizon be N_p . The predicted state trajectory is generated according to

$$\hat{s}_{t+k+1} = f_\theta(\hat{s}_{t+k}, u_{t+k}, \tilde{w}_{t+k}), \quad \hat{s}_t = s_t \quad k = 0, \dots, N_p - 1 \quad (21)$$

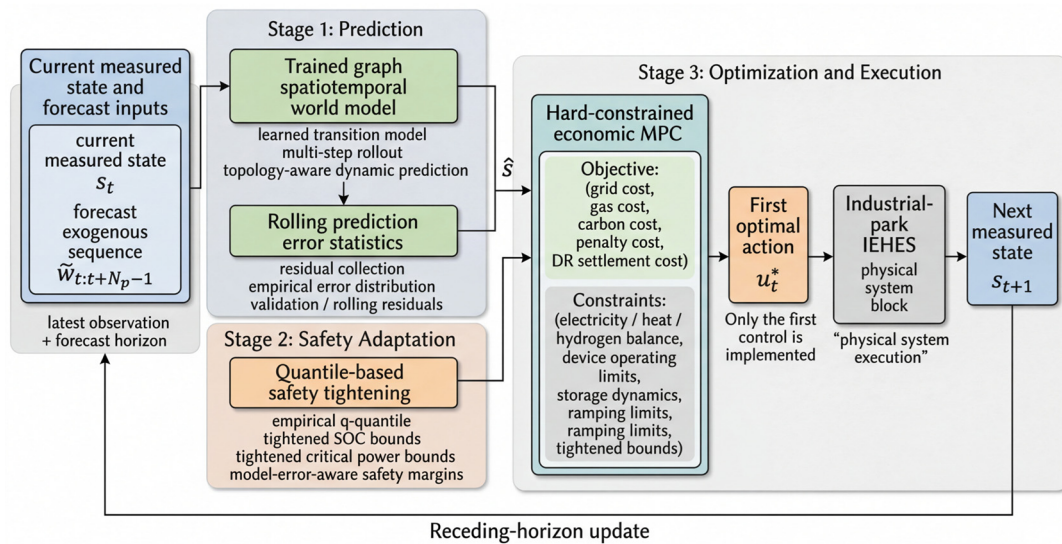


Figure 3. Rolling MPC framework embedding the learned world model and quantile-based tightened constraints for closed-loop dispatch.

Equation (21) defines the internal predictive kernel of the MPC layer. At each control step, the future trajectory generation starts from the current measured state. The forecast exogenous sequence, including renewable availability, electricity/heat/hydrogen demand, and market-related signals, is provided over the MPC horizon. For a candidate control sequence evaluated by the MPC optimizer, the current state is first mapped into node-level graph features together with the corresponding control and exogenous inputs. The graph encoder extracts topology-aware electricity–heat–hydrogen coupling representations, while the temporal encoder summarizes the recent historical operating window. The decoder then predicts the next state. This predicted state is appended to the rolling input window and recursively used to generate the following state until the end of the prediction horizon. Therefore, the MPC optimizer evaluates each candidate control sequence based on the future state trajectory recursively generated by the graph spatiotemporal world model. In this way, the dispatch optimizer plans over a learned transition model rather than a fixed mechanistic approximation, while still preserving an explicit optimization structure for economic dispatch under hard constraints.

4.2. Learned-Dynamics-Constrained Economic MPC Formulation

Given the learned transition model in Equation (21), the receding-horizon dispatch problem at time t is formulated as a finite-horizon constrained optimization problem. Let the decision variable be the control sequence

$$U_t = \{u_\tau \in \mathcal{U}\}_{\tau=t}^{t+N_p-1} \quad (22)$$

The objective is to minimize the cumulative operating cost over the prediction horizon while discouraging excessive control variation. The economic MPC objective is written as

$$\min_{U_t} J_t(U_t) = \sum_{\tau=t}^{t+N_p-1} (C_\tau^{\text{grid}} + C_\tau^{\text{gas}} + C_\tau^{\text{carbon}} + C_\tau^{\text{penalty}} + C_\tau^{\text{dr}} + \rho \|\Delta u_r\|_2^2) \quad (23)$$

where $\rho \geq 0$ is a regularization coefficient used to suppress excessive step-to-step control variation.

To make the role of the learned transition dynamics explicit, the proposed MPC problem solved at each time step is written as a learned-dynamics-constrained finite-horizon

optimization problem. The decision variable is the control sequence $U_t = \{u_\tau\}_{\tau=t}^{t+N_p-1}$, while the predicted state sequence is recursively generated by the learned graph spatiotemporal world model. The complete MPC formulation is given by

$$\begin{aligned} \min_{U_t, \hat{S}_t} J_t(U_t) &= \sum_{\tau=t}^{t+N_p-1} (C_\tau^{grid} + C_\tau^{gas} + C_\tau^{carbon} + C_\tau^{penalty} + C_\tau^{dr} + \rho \|\Delta u_r\|_2^2) \\ \text{s.t.} \quad &\hat{S}_t = S_t, \\ &\hat{S}_{\tau+1} = f_\theta(\hat{S}_\tau, u_\tau, w_\tau), \\ &\phi^{el}(\hat{S}_\tau, u_\tau, w_\tau) = 0, \\ &\phi^{th}(\hat{S}_\tau, u_\tau, w_\tau) = 0, \\ &\phi^{h2}(\hat{S}_\tau, u_\tau, w_\tau) = 0, \\ &\hat{S}_t \in S_\tau^{tight}, u_\tau \in \mathcal{U}_\tau^{tight}, \\ &|u_\tau - u_{\tau-1}| \leq \Delta r, \\ &u_\tau \in \mathcal{U}_\tau^{DR} \end{aligned} \quad (24)$$

where $\hat{S}_t = \{\hat{S}_\tau\}_{\tau=t}^{t+N_p}$ denotes the predicted state trajectory, $f_\theta(\cdot)$ is the learned graph spatiotemporal transition model, and $S_\tau^{tight}, \mathcal{U}_\tau^{tight}$ denote the tightened state and control feasible sets. The equality constraint $\hat{S}_{\tau+1} = f_\theta(\hat{S}_\tau, u_\tau, w_\tau)$ is the key difference between the proposed formulation and a conventional economic MPC with a fixed nominal model. It forces every candidate control sequence to be evaluated through the learned multi-energy transition dynamics, so that the predicted SOC, TES state, hydrogen inventory, and critical power variables evolve consistently with the topology-aware world-model rollout.

Compared with classical MPC, the proposed formulation retains the standard receding-horizon structure, economic objective, and explicit operational constraints, but replaces the fixed nominal transition model with the learned graph spatiotemporal transition model f_θ . In classical MPC, the predicted trajectory is usually generated by simplified mechanistic equations or linearized nominal dynamics. Such a formulation is transparent but may be inaccurate when the IEHES exhibits strong electricity–heat–hydrogen coupling, storage carry-over effects, and forecast-driven operating variations. In the proposed MPC, the learned transition constraint allows the optimizer to evaluate control sequences under data-driven multi-step dynamics that have been trained with topology-aware message passing and constraint-aligned rollout learning. Therefore, the advantage of embedding the learned transition dynamics is not that the hard constraints are removed, but that the internal prediction model used by the constrained optimizer becomes more adaptive to coupled multi-energy dynamics. The quantile-tightened bounds are then imposed on the predicted trajectory to reduce the risk that residual rollout errors lead to violations of the original physical limits.

Although the world model is trained using both prediction loss and physics-consistency regularization, its rollout error cannot be assumed to vanish in online operation. When the learned transition model is embedded into the MPC as the internal prediction kernel, residual prediction errors may accumulate over the horizon and cause nominally feasible trajectories to violate hard constraints in the actual system. The uncertainty propagation considered here is therefore trajectory-dependent. At each prediction step, the state predicted by the world model becomes part of the input for the next rollout step. Consequently, a residual error in battery SOC, TES state, hydrogen inventory, or a critical power variable can affect the subsequent feasible charging/discharging range, conversion output, and multi-energy balance. Through CHP electricity–heat coupling, electrolyzer electricity–hydrogen conversion, and storage carry-over dynamics, local prediction errors may propagate across energy carriers and across time steps. As a result, a control sequence

that is feasible under the nominal predicted trajectory may become infeasible during online execution if the predicted trajectory operates too close to the original physical limits. To address this issue, the proposed method tightens key state and power constraints according to empirical quantiles of the model's rolling prediction errors.

Let e_t^{soc} denote the one-step or multi-step prediction error of a storage-related state, and let $e_t^{p,i}$ denote the prediction error of the i -th critical power-related quantity. Based on a validation set or a sliding window of rolling closed-loop residuals, the empirical q -quantile of the absolute prediction error is computed. The resulting tightening margins are defined as

$$\delta_{SOC} = Q_q(|e_t^{soc}|) \quad (25)$$

$$\delta_{p,i} = Q_q(|e_t^{p,i}|) \quad (26)$$

where $Q_q(\cdot)$ denotes the empirical q -quantile operator. Accordingly, the core problem addressed in this paper is a finite-horizon constrained economic dispatch problem over learned system dynamics.

In the implementation, the quantile level was set to $q = 0.95$. The tightening margins were calibrated using the validation segment rather than the final test segment. For each selected variable, N_{cal} rolling validation samples were used, and each sample produced a 24-step rollout error sequence; therefore, the empirical 95% quantile was calculated from $24N_{cal}$ absolute prediction-error values. The margins were computed separately for battery SOC, TES state, hydrogen inventory, grid import, CHP output, gas-boiler output, electrolyzer power, and storage charging/discharging power. Thus, the MPC did not use a uniform tightening coefficient, but adopted variable-specific margins obtained from the corresponding empirical prediction-error distributions.

The state of charge bounds used in the MPC are then tightened from the nominal interval $[SOC, \overline{SOC}]$ to

$$\underline{SOC}' = \underline{SOC} + \delta_{SOC}, \quad \overline{SOC}' = \overline{SOC} - \delta_{SOC} \quad (27)$$

and the power bounds of the i -th critical variable are similarly tightened as:

$$\underline{p}'_i = \underline{p} + \delta_{p,i}, \quad \overline{p}'_i = \overline{p} - \delta_{p,i} \quad (28)$$

In practical implementation, the tightening margins are obtained from the rolling prediction residuals on the validation set. For a selected storage-related state, such as battery SOC, TES state, or hydrogen inventory, the absolute k -step prediction error is first calculated as the difference between the predicted value and the measured value over all validation samples. The empirical α -quantile of these absolute errors is then used as the tightening margin. The same procedure is applied to critical power-related variables, such as grid import, CHP output, electrolyzer power, or storage charging/discharging power. During MPC optimization, the nominal lower bound is increased by the corresponding quantile margin, while the nominal upper bound is decreased by the same margin. Therefore, the optimizer does not plan directly on the original physical boundary, but on a slightly narrower feasible interval. If the learned trajectory contains residual prediction errors during online execution, the tightened interval provides a buffer that reduces the probability of violating the original hard constraints. For example, if the nominal battery SOC bound is $[0.10, 0.90]$ and the 95% empirical quantile of the rolling SOC prediction error is 0.02, the tightened SOC interval used by MPC becomes $[0.12, 0.88]$. In this way, the empirical prediction-error distribution is directly transformed into an executable safety margin. Unlike fixed safety margins selected heuristically, the tightening magnitudes are explicitly linked to the empirical error distribution of the learned world model. Therefore, the

resulting margins are interpretable, data-dependent, and adjustable through the quantile level q .

4.3. Tightened MPC Problem and Online Rolling Procedure

To make the comparison between non-tightened and tightened MPC clearer, the non-tightened MPC considered in this study refers to the same world-model-driven economic MPC solved with the original nominal state and power bounds. In this case, the learned transition model, economic objective, multi-energy balance constraints, device operating limits, ramping constraints, and DR contractual constraints are all kept unchanged, but no explicit prediction-error buffer is reserved around critical variables. By contrast, the tightened MPC uses the same objective function and learned world model, while replacing the nominal SOC and critical power bounds with the quantile-tightened bounds defined in Equations (27) and (28). Therefore, the difference between the two formulations lies only in the feasible region used by the optimizer. This constraint-tightening idea is consistent with robust and learning-based MPC studies, where safety margins are introduced to reduce the risk of constraint violation under model uncertainty or prediction residuals [27].

After introducing quantile-based safety tightening, the constrained optimization problem solved at each time step is modified by replacing the nominal box constraints with the tightened bounds in Equations (29) and (30) for the selected state and power variables. The resulting optimization problem can be written as

$$\min_{U_t} J_t(U_t) \quad (29)$$

subject to the learned transition dynamics in Equation (21), the multi-energy balance constraints in Equation (7), the device operating and ramping constraints in Equations (8)–(10), and the tightened bounds defined in Equations (27) and (28).

$$\underline{s}' \leq \hat{s}_\tau \leq \bar{s}', \underline{u}' \leq \hat{u}_\tau \leq \bar{u}' \quad (30)$$

where $\underline{s}'$, $\bar{s}'$, $\underline{u}'$, and $\bar{u}'$ denote the tightened lower and upper bounds derived from the empirical prediction-error quantiles. In practice, the tightening is applied only to those variables whose boundary violation is critical for safe operation, such as battery SOC and selected power variables directly influencing balance feasibility or ramping saturation. This selective tightening prevents excessive conservatism in variables that are less sensitive to model residuals while still providing an explicit guard against model-induced boundary crossing.

After the tightening margins are obtained, the nominal state and power bounds in the MPC problem are replaced by their tightened counterparts for selected critical variables. The control sequence remains the decision variable, while the future state trajectory is recursively generated by the learned world model under each candidate control sequence. The tightened bounds are imposed directly as hard constraints inside the finite-horizon MPC rather than as a post-processing correction. Therefore, any candidate control sequence that causes the predicted SOC or critical power variables to exceed the tightened feasible region is regarded as infeasible. By reserving an explicit buffer between the optimized trajectory and the original physical limits, the tightened MPC improves closed-loop feasibility under residual rollout errors without changing the economic objective.

The improvement brought by quantile tightening lies in the modification of the feasible region used by the MPC optimizer. Instead of allowing the predicted trajectory to operate directly on the nominal physical limits, the tightened formulation reserves an error buffer between the optimized trajectory and the original hard constraints. Therefore, even when residual rollout errors occur during online execution, the implemented trajectory is less

likely to cross the true SOC or power bounds. In this sense, the tightened MPC formulation improves closed-loop feasibility not by changing the economic objective, but by making the admissible control sequences more conservative around critical operating boundaries.

5. Experimental Results

5.1. Case Study and Evaluation Protocol

The proposed framework is validated on an industrial-park integrated electricity–heat–hydrogen energy system over a monthly evaluation horizon with an hourly dispatch resolution. The case study considers coupled electric, thermal, and hydrogen demands, together with renewable energy availability from photovoltaic and wind sources. As shown in Figure 4, the electricity, thermal, and hydrogen demand trajectories exhibit evident multi-timescale variations across the month. Meanwhile, Figure 5 presents the corresponding photovoltaic and wind power availability, highlighting the strong diurnal intermittency of solar generation and the continuous stochastic fluctuation of wind power.

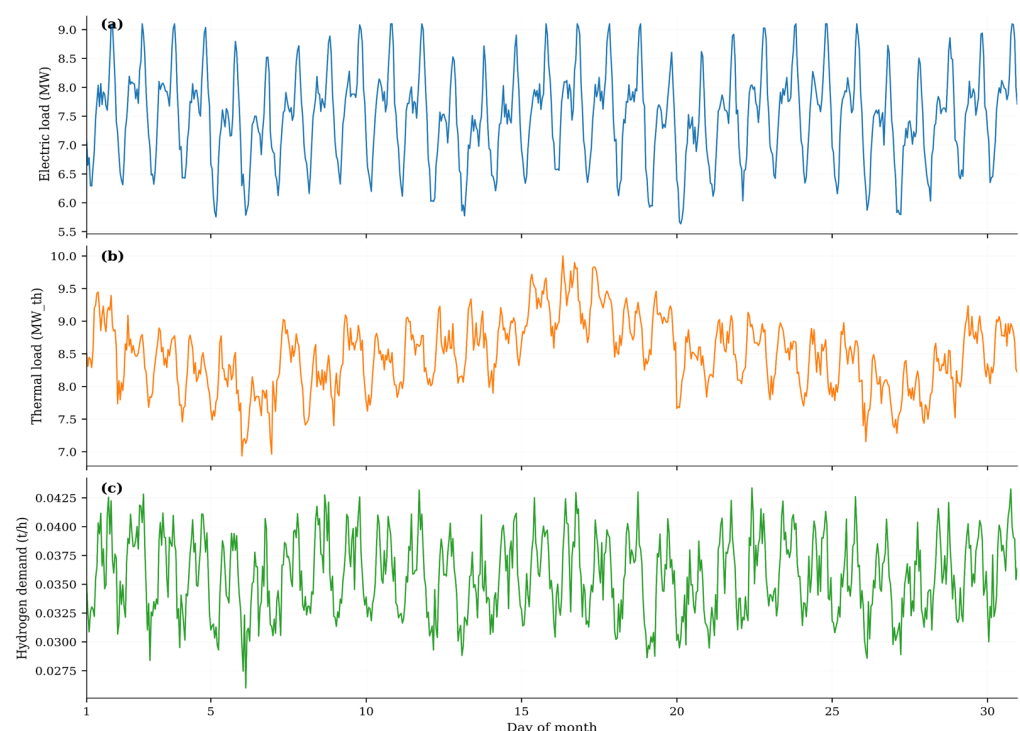


Figure 4. The monthly electricity, heat, and hydrogen demand profiles in the held-out evaluation horizon. (a) electric load profile, showing the electrical power demand variation over the month; (b) thermal load profile, showing the thermal power demand variation over the month; (c) hydrogen demand profile, showing the hydrogen consumption requirement over the month.

The dataset used in this study consists of hourly operational time-series data of the industrial-park IEHES. The input variables include photovoltaic and wind power availability, electricity load, thermal load, hydrogen demand, electricity price, gas price, and carbon-related price signals. The output and state variables include grid import, CHP electric and thermal outputs, gas-boiler output, electrolyzer power, battery SOC, TES state, hydrogen inventory, and storage charging/discharging quantities. To avoid information leakage in time-series prediction and rolling dispatch evaluation, the dataset was partitioned chronologically rather than randomly. Specifically, the earlier part of the historical data was used for world model training, the subsequent validation segment was used for hyperparameter selection, early stopping, and empirical prediction-error quantile estimation, and the final held-out monthly segment was used only for predictive

accuracy evaluation and closed-loop dispatch comparison. All normalization parameters were calculated from the training set and then applied unchanged to the validation and test sets. Rolling samples were constructed using a 24 h historical input window and a 24-step prediction horizon, which is consistent with the MPC horizon adopted in the dispatch experiments.

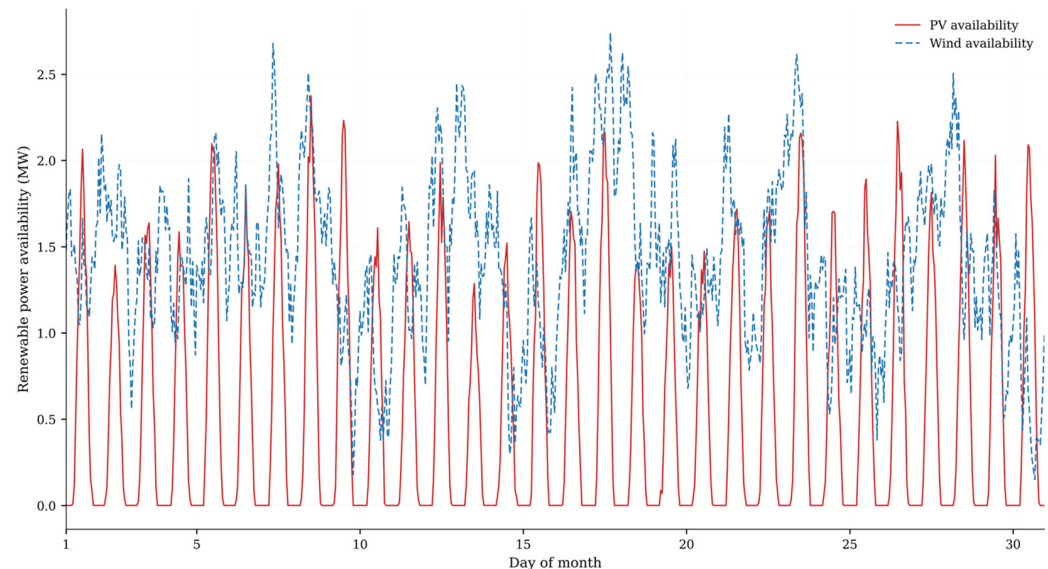


Figure 5. The monthly PV and wind availability profiles in the held-out evaluation horizon.

To make the experimental data description more complete, the dataset used in this study is further clarified as follows. The raw records are hourly time-series data collected from the studied industrial-park IEHES, including renewable availability, electricity/heat/hydrogen demand, market-related price signals, device operating outputs, and storage states. The dataset was chronologically divided into training, validation, and held-out test segments. The training segment was used to learn the graph spatiotemporal world model; the validation segment was used for hyperparameter selection, early stopping, and empirical prediction-error quantile estimation; and the held-out monthly test segment was used only for prediction evaluation and closed-loop dispatch comparison. All normalization parameters were fitted using the training segment only and then applied unchanged to the validation and test segments. In addition, the perturbation tests were generated by applying $\pm 5\%$, $\pm 10\%$, and $\pm 15\%$ disturbances to renewable and demand inputs, while keeping the same MPC settings and device constraints across all compared methods.

Before model training, the hourly time-series data were pre-processed in a unified manner. Missing or abnormal entries were first checked according to device operating ranges and temporal continuity. When isolated missing values occurred, linear interpolation was used; abnormal points outside the physical operating bounds were clipped to the corresponding admissible range. Continuous variables with different units and magnitudes, such as MW, MW_{th}, SOC, hydrogen inventory, and price signals, were normalized using statistics fitted only on the training set. The same pre-processing and normalization procedure was applied to all benchmark predictors to ensure a fair comparison. For reproducibility, the historical window length, prediction horizon, node-feature construction, optimizer, learning rate, batch size, number of epochs, gradient clipping threshold, and loss weights are reported in Table 3. All the models were trained and tested under the same chronological data split, input variables, prediction horizon, and MPC settings, so that the reported differences mainly reflect the modeling and dispatch mechanisms rather than inconsistent experimental settings.

The evaluation protocol is designed from two complementary perspectives. First, the predictive capability of the world model is assessed in terms of multi-step state-transition accuracy, which provides the basis for closed-loop receding-horizon dispatch. Second, the operational performance of the overall framework is examined from the viewpoints of economic efficiency, low-carbon operation, and dispatch feasibility over the monthly horizon. In addition to the month-scale assessment, representative days are further extracted for a more detailed analysis of intra-day operating behaviors. This protocol enables a unified evaluation of both model quality and control effectiveness under realistic time-varying operating conditions.

5.2. Predictive Accuracy of the World Model

To evaluate whether the learned dynamics are sufficiently accurate for downstream rolling optimization, the proposed GraphWorldModel_MPC was tested on the held-out monthly horizon using the same 24-step rollout length as that adopted in both training and MPC. Table 4 reports representative one-step and 24-step prediction errors for key dispatch-related states. Overall, the learned world model maintained high predictive fidelity across both power-side variables and storage-related states, with the average 24-step RMSE remaining below 5% of the corresponding variable range. This indicates that the model preserved the dominant temporal structure of the industrial-park IEHES over the full receding-horizon window rather than only under one-step prediction.

Table 4. Representative prediction accuracy of the proposed world model on key dispatch-related states.

Variable	1-Step MAE	1-Step RMSE	24-Step MAE	24-Step RMSE	24-Step R^2
Grid import (MW)	0.068	0.090	0.178	0.220	0.938
CHP electric output (MW)	0.022	0.028	0.047	0.060	0.924
CHP heat output (MW_th)	0.030	0.037	0.063	0.080	0.926
Gas-boiler output (MW_th)	0.035	0.045	0.079	0.095	0.925
Electrolyzer power (MW)	0.024	0.031	0.058	0.070	0.927
Battery SOC (-)	0.006	0.008	0.011	0.015	0.997
TES state (-)	0.009	0.011	0.018	0.022	0.980
H2 inventory (t)	0.0026	0.0035	0.0062	0.0080	0.995

The prediction accuracy was especially strong for storage-related states. The 24-step R^2 values of battery SOC, TES state, and hydrogen inventory reached 0.997, 0.980, and 0.995, respectively, while their corresponding 24-step RMSEs remained limited to 0.015, 0.022, and 0.008 in their own units. Such results suggest that the proposed graph spatiotemporal architecture effectively captured the intertemporal carry-over effects induced by charging/discharging dynamics and multi-energy conversion coupling. This is particularly important for the downstream rolling dispatch problem, because storage-state drift is one of the main sources of recursive prediction accumulation in long-horizon MPC. For power-related variables, the learned model also achieved satisfactory multi-step accuracy. Even at the end of the 24-step rollout, the RMSEs remained within 0.22 MW for grid import, 0.06 MW for CHP electric output, 0.08 MW_th for CHP heat output, 0.095 MW_th for gas-boiler output, and 0.07 MW for electrolyzer power. Considering the operating ranges observed in the monthly evaluation set, these errors remain moderate and do not alter the dominant dispatch pattern. Moreover, the prediction error increased gradually with the rollout horizon instead of exhibiting abrupt divergence, indicating that the physics-consistency regularization introduced in Section 3 effectively improved the long-horizon stability of the learned dynamics.

Figure 6 presents the actual trajectories together with the 1-step and 24-step predictions for four representative variables on a typical day, including grid import, CHP electric output, battery SOC, and hydrogen inventory. Overall, the proposed world model preserves the main temporal evolution patterns over the full 24-step rollout horizon. The 1-step predictions almost overlap with the actual trajectories for all variables, while the 24-step predictions still track the dominant trends with only moderate deviations. For the power-related variables, including grid import and CHP electric output, the 24-step prediction errors become slightly more visible around local peaks, valleys, and transition intervals, but the model still captures the overall dispatch pattern well. For the storage-related states, the prediction accuracy is even higher; the battery SOC trajectory, including the upper-bound saturation period and the subsequent discharge process, is reproduced almost exactly, and the hydrogen inventory prediction also remains closely aligned with the actual state evolution throughout the day. These results indicate that the learned world model maintains sufficiently high multi-step predictive fidelity to support the downstream receding-horizon dispatch optimization.

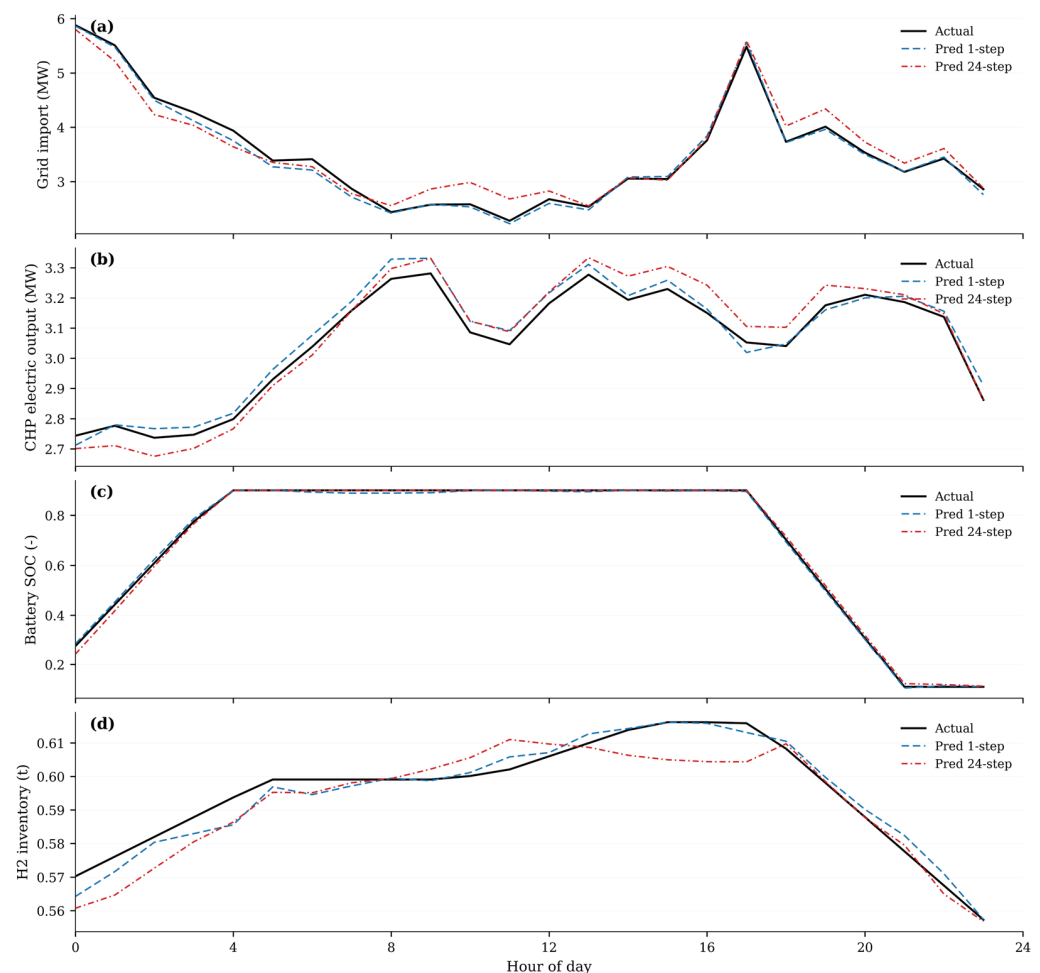


Figure 6. Actual, 1-step predicted, and 24-step predicted trajectories of representative dispatch-related variables. (a) Grid import power; (b) CHP electric output; (c) battery state of charge; (d) hydrogen storage inventory.

5.3. Representative-Day and Monthly Operational Results

Figures 7 and 8 illustrate the representative-day operating behaviors of the proposed GraphWorldModel_MPC. As shown in Figure 7, the electricity-, heat-, and hydrogen-side balances are satisfied on an hourly basis, and the positive and negative stacked components remain well matched throughout the day, indicating that the obtained dispatch

solution preserves multi-energy feasibility under the coupled operating constraints. On the electricity side, the adjusted electric load is jointly supported by grid import, CHP electric output, renewable generation, and battery discharge, while part of the electricity is simultaneously allocated to the electrolyzer and battery charging during selected hours. The CHP electric output remains relatively stable, providing a firm local supply backbone, whereas grid import exhibits stronger time variation and serves as the main balancing component in response to load fluctuations and renewable intermittency.



Figure 7. Representative-day electricity, heat, and hydrogen balance results obtained by Graph-WorldModel_MPC. (a) Electricity-side source-load-storage balance; (b) heat-side source-load-storage balance; (c) hydrogen-side production-demand-storage balance. Positive bars denote demand or charging components, whereas negative bars denote supply or discharging components.

On the heat side, the thermal demand is mainly met by the coordinated contribution of CHP heat output and the gas-fired boiler, while TES is used as an additional temporal buffer. The CHP heat output stays at a relatively stable level over the day, whereas the gas-boiler provides the main flexibility to absorb residual thermal imbalance. Around the middle of the day, TES charging becomes active, indicating that part of the available thermal supply is shifted forward for later use. During the subsequent peak-demand period, TES discharge is activated to reduce the instantaneous dependence on direct thermal generation. This behavior confirms that the proposed dispatch strategy can exploit thermal storage to reshape intra-day heat supply without violating the heat-balance constraints. The hydrogen-side behavior exhibits a similar time-shifting pattern. As shown in Figure 7c, hydrogen demand is met by the combined effect of electrolyzer-based hydrogen production and hydrogen tank discharge, while hydrogen tank charging is activated when production exceeds immediate demand. Correspondingly, Figure 8c shows that the hydrogen inventory first increases and then decreases, reflecting a clear store-and-release pattern across the day.

This indicates that the dispatch framework effectively coordinates electricity-to-hydrogen conversion with hydrogen storage to decouple, to some extent, instantaneous production from end-use demand.

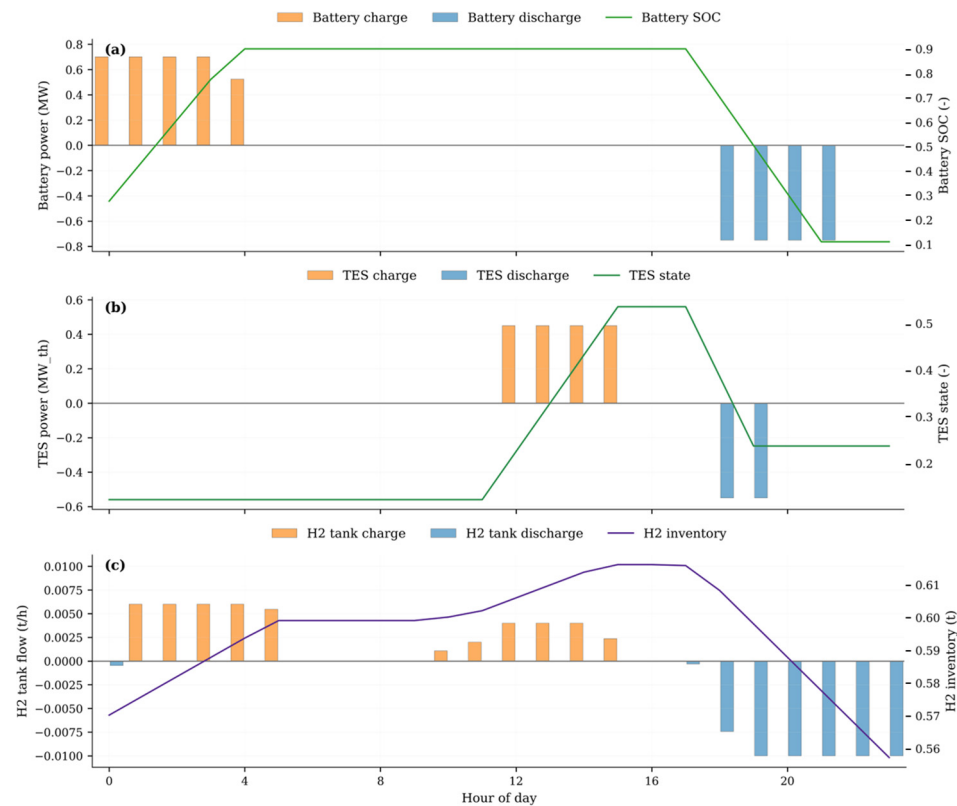


Figure 8. Representative-day charging/discharging actions and state trajectories of battery, thermal, and hydrogen storage. (a) Battery charge/discharge power and SOC trajectory; (b) TES charge/discharge power and thermal storage state; (c) hydrogen tank charge/discharge flow and hydrogen inventory trajectory.

Figure 8 further reveals the intertemporal roles of the three storage resources. The battery is charged during the early hours and discharged during the evening period, with its state of charge rising rapidly to the upper bound and later decreasing to the lower bound. This behavior suggests that the battery is mainly used for short-term electricity-side arbitrage and peak support. By contrast, TES is charged around midday and discharged over a narrower evening window, highlighting its role in smoothing thermal mismatch across adjacent hours. The hydrogen tank exhibits a longer-duration inventory cycle, with charging occurring in both the early and midday periods and discharging concentrated in the later hours. Taken together, these results show that the proposed framework allocates battery, TES, and hydrogen storage to distinct yet coordinated temporal functions, thereby improving the flexibility of the integrated electricity–heat–hydrogen system.

Overall, the representative-day results demonstrate that the proposed method can simultaneously maintain strict multi-energy balance and exploit cross-carrier storage flexibility to realize coordinated intra-day dispatch. The operational patterns observed in Figures 7 and 8 also provide a physical explanation for the favorable month-scale performance reported later, namely that the learned world-model-driven rolling MPC can transform short-term renewable availability and storage flexibility into feasible and economically effective multi-energy scheduling decisions.

5.4. Comparative Dispatch Performance

Conventional EMPC denotes a deterministic economic MPC method, in which the future state trajectory is generated by a nominal predictive model and the economic objective is optimized over the receding horizon. DRAMPC denotes a distributionally robust adaptive MPC method, which incorporates uncertainty-aware adjustment into the rolling optimization process and is therefore more conservative than deterministic EMPC under forecast mismatch. GRU-MPC replaces the nominal predictive model with a recurrent neural network predictor, but it treats the IEHES state as a sequence-only representation and does not explicitly encode the directed electricity–heat–hydrogen topology or physics-consistency constraints. These benchmark methods were implemented by following the model structures, parameter settings, and tuning principles reported in their corresponding reference papers as closely as possible. For parameters that were not explicitly specified in the original references or had to be adapted to the studied IEHES scale, the validation segment was used for selection, and the held-out monthly test segment was not used for tuning.

For fairness, all compared methods were evaluated under the same industrial-park IEHES dataset, hourly resolution, 24-step MPC horizon, device operating constraints, multi-energy balance constraints, cost-function components, and evaluation metrics. Therefore, the comparison does not rely on arbitrarily weakened baselines, but follows a reference-guided reproduction protocol under a unified dispatch environment. By contrast, the proposed GraphWorldModel_MPC uses a topology-aware graph spatiotemporal world model with multi-step rollout learning and physics-consistency regularization, and further embeds quantile-based safety tightening into the hard-constrained economic MPC. Therefore, although all compared methods follow the economic MPC principle, they differ in the internal prediction model, uncertainty-handling mechanism, and constraint-tightening strategy.

As shown in Table 5, the proposed method achieves the lowest total operating cost of 5,757,859 RMB, outperforming Conventional EMPC, DRAMPC, and GRU-MPC by 6.07%, 3.83%, and 10.79%, respectively. This advantage can be explained by the different predictive and constraint-handling mechanisms of the compared MPC formulations. Conventional EMPC relies on nominal dynamics and therefore has limited adaptability when renewable generation, load demand, and storage states evolve under coupled electricity–heat–hydrogen interactions. As a result, it yields higher grid import, gas cost, carbon emissions, and penalty cost than the proposed method.

Table 5. Monthly comparative dispatch performance of different methods.

Method	Total Cost (RMB)	Grid Cost (RMB)	Gas Cost (RMB)	Carbon Cost (RMB)	Penalty Cost (RMB)	DR Settlement Cost (RMB)	Grid Import (MWh)	Equivalent Carbon Emissions (tCO ₂)
Conventional EMPC [28]	6,128,743	2,086,417	3,692,581	324,904	11,286	13,555	3223.84	4061.30
DRAMPC [12]	5,987,416	2,022,864	3,603,975	316,842	6731	37,004	3147.62	3960.53
GRU-MPC [26]	6,453,918	2,194,376	3,818,925	334,278	87,461	18,878	3311.47	4178.48
Ours	5,757,859	1,949,522	3,476,646	302,519	2479	26,693	3060.29	3781.49

DRAMPC improves robustness by considering uncertainty in the rolling optimization process, and therefore performs better than Conventional EMPC in terms of total cost and penalty cost. However, its robust treatment tends to be conservative and does not explicitly learn the structured spatiotemporal dynamics of the IEHES. Consequently, although DRAMPC reduces constraint-related penalties relative to Conventional EMPC, it still produces higher grid import, gas cost, and carbon cost than GraphWorldModel_MPC.

GRU-MPC introduces a learning-based predictive model and can capture part of the temporal dependency in the operational data. Nevertheless, its sequence-only structure ignores the directed topology and cross-carrier coupling paths among the electricity,

heat, and hydrogen subsystems. This limitation leads to less accurate multi-step rollout trajectories and weaker constraint satisfaction during closed-loop dispatch, as reflected by its highest total cost, highest carbon emissions, and largest penalty cost among the compared methods.

By contrast, the proposed GraphWorldModel_MPC combines topology-aware graph encoding, temporal rollout prediction, physics-consistency regularization, and quantile-based safety tightening. These components improve the quality of predicted trajectories and reduce the probability of constraint violation during MPC execution. Therefore, the proposed method achieves lower grid import, gas consumption, carbon emissions, and penalty cost simultaneously. The results indicate that the performance gain does not come merely from using an economic MPC objective, but from improving the predictive kernel and feasibility-preserving mechanism embedded inside the MPC formulation.

5.5. Robustness, Ablation, and Sensitivity

To investigate the robustness of the proposed framework under forecast uncertainty, multi-source uncertainty tests were conducted over the monthly evaluation horizon. In the implementation, forecast errors were introduced into photovoltaic output, wind output, electric load, thermal load, and hydrogen demand, respectively. For each uncertainty level, namely $\pm 5\%$, $\pm 10\%$, and $\pm 15\%$, random relative error factors were independently generated within the corresponding uncertainty range and applied to the above renewable and demand inputs. The rolling MPC dispatch was then repeatedly evaluated under the generated uncertainty realizations, and the results reported in Table 6 are the averaged values over these repeated tests. Therefore, the reported robustness results reflect the average closed-loop dispatch performance under joint renewable-side and multi-energy demand-side uncertainty, rather than a single deterministic perturbation case.

Table 6. Robustness comparison under different perturbation levels.

Method	Perturbation Level	Total Cost (RMB)	Increase vs. Nominal	Penalty Cost (RMB)	Equivalent Carbon Emissions (tCO ₂)	Dispatch Violation Rate
DRAMPC	$\pm 5\%$	6,094,287	1.79%	8462	3989.74	0.21%
DRAMPC	$\pm 10\%$	6,199,974	3.55%	12,638	4027.31	0.57%
DRAMPC	$\pm 15\%$	6,337,125	5.84%	19,486	4081.66	1.08%
GRU-MPC	$\pm 5\%$	6,590,864	2.12%	96,315	4214.57	1.46%
GRU-MPC	$\pm 10\%$	6,752,109	4.62%	118,782	4286.39	2.84%
GRU-MPC	$\pm 15\%$	6,938,471	7.51%	146,225	4365.08	4.73%
GraphWorldModel_MPC	$\pm 5\%$	5,833,944	1.32%	3108	3808.66	0.00%
GraphWorldModel_MPC	$\pm 10\%$	5,921,476	2.84%	4287	3846.92	0.00%
GraphWorldModel_MPC	$\pm 15\%$	6,018,023	4.52%	6914	3895.37	0.14%

Table 5 shows that the proposed GraphWorldModel_MPC consistently exhibits the smallest performance degradation among the compared methods. As the uncertainty level increases from $\pm 5\%$ to $\pm 15\%$, the monthly total cost of the proposed method rises from 5.834 million RMB to 6.018 million RMB, corresponding to a relative increase of 1.32–4.52% over the nominal case, while the penalty cost remains limited and the dispatch violation rate stays close to zero. By contrast, DRAMPC shows a larger cost increase and a higher violation rate under stronger uncertainty, and GRU-MPC degrades more visibly, especially at the $\pm 15\%$ level. These results indicate that the proposed framework preserves stronger closed-loop robustness against multi-source forecast mismatch, which can be attributed to its more accurate multi-step state prediction, learned transition dynamics embedded in the MPC, and quantile-based safety tightening.

After the robustness tests, ablation studies were further conducted to examine the contribution of the main modeling components. Specifically, the graph encoder in the world model is replaced by a sequence-only recurrent predictor with the same MPC

layer, denoted as SeqWorldModel_MPC. As reported in Table 7, removing the graph representation increases the monthly total cost from 5.758 million RMB to 5.913 million RMB, together with higher grid import, higher equivalent carbon emissions, and a larger penalty cost. In addition, removing temporal encoding further increases the total cost to 5.946 million RMB and the penalty cost to 5936 RMB, confirming that the historical trajectory information is necessary for capturing storage carry-over effects, ramping history, and lagged exogenous variations. When the multi-step rollout training is removed, the total cost also increases to 5.889 million RMB, indicating that one-step training alone is insufficient to match the recursive prediction mode used in MPC. Overall, these results demonstrate that the graph representation, temporal encoding, and multi-step rollout training jointly improve the dynamic representation quality of the world model and lead to more effective rolling dispatch decisions. In addition, the model size and rollout horizon remain moderate under the adopted settings in Table 3, so the additional computational burden introduced by graph encoding and temporal rollout is acceptable for hourly rolling dispatch.

Table 7. The ablation and sensitivity results of the proposed framework.

Setting	Total Cost (RMB)	Grid Import (MWh)	Equivalent Carbon Emissions (tCO ₂)	Penalty Cost (RMB)
SeqWorldModel_MPC (<i>w/o</i> graph representation)	5,912,684	3118.73	3870.44	3516
GraphWorldModel_MPC	5,757,859	3060.29	3781.49	2479
GraphWorldModel_MPC, horizon = 12	5,821,436	3087.16	3804.27	2843
GraphWorldModel_MPC, horizon = 24	5,757,859	3060.29	3781.49	2479
GraphWorldModel_MPC, horizon = 36	5,783,904	3069.58	3792.16	2761
GraphWorldModel_MPC <i>w/o</i> temporal encoding	5,946,372	3129.84	3884.61	5936
GraphWorldModel_MPC <i>w/o</i> multi-step rollout training	5,889,246	3104.57	3842.36	4718

5.6. Role of Constraint-Aligned Training and Quantile Tightening

To quantify the practical contribution of constraint-aligned training and quantile-based safety tightening within the proposed framework, two degraded variants were further tested. The first variant, GraphWorldModel_MPC *w/o* physics-consistency, removes the balance-, storage-boundary-, and ramping-consistency terms from world model training. This case is used only as an ablation setting to evaluate the effect of constraint-aligned training, rather than as a practical dispatch configuration. The second variant, GraphWorldModel_MPC *w/o* quantile tightening, solves the MPC problem using the nominal state and power bounds without uncertainty-aware tightening. The comparison results are summarized in Table 8.

Table 8. Effects of constraint-aligned training and quantile-based safety tightening.

Setting	Avg. 24-Step NRMSE (%)	Total Cost (RMB)	Grid Import (MWh)	Equivalent Carbon Emissions (tCO ₂)	Penalty Cost (RMB)	Dispatch Violation Rate
GraphWorldModel_MPC <i>w/o</i> physics-consistency	6.11	5,868,417	3101.74	3836.92	8236	0.63%
GraphWorldModel_MPC <i>w/o</i> quantile tightening	4.31	5,811,904	3066.38	3789.47	14,582	0.97%
GraphWorldModel_MPC	4.28	5,757,859	3060.29	3781.49	2479	0.00%

Removing the physics-consistency regularization leads to a clear deterioration in both predictive quality and dispatch performance. As shown in Table 8, the average 24-step NRMSE increases from 4.28% to 6.11%, while the monthly total cost and penalty cost rise from 5.758 million RMB and 2479 RMB to 5.868 million RMB and 8236 RMB, respectively. This indicates that the constraint-aligned training terms help guide the learned rollout trajectories toward the physical feasible manifold of the IEHES, thereby improving both prediction stability and downstream dispatch performance.

The comparison between GraphWorldModel_MPC *w/o* quantile tightening and the complete GraphWorldModel_MPC directly reflects the effect of non-tightened versus tightened MPC. Both settings use the same predictive model and economic objective, but the non-tightened variant uses nominal state and power bounds without prediction-error margins. Although its prediction accuracy remains almost unchanged, with the average 24-step NRMSE changing only from 4.28% to 4.31%, the penalty cost increases from 2479 RMB to 14,582 RMB and the dispatch violation rate rises from 0.00% to 0.97%. This confirms that quantile-based tightening mainly improves closed-loop feasibility by reserving executable safety margins inside the MPC feasible region, rather than by improving nominal prediction accuracy.

5.7. Comparison with Advanced Time-Series Prediction Baselines

To position the proposed predictor relative to advanced time-series forecasting methods, additional experiments were conducted under a unified predictor-to-MPC interface. The historical window, prediction horizon, input variables, and downstream hard-constrained MPC were kept identical for all methods, while only the internal predictor was replaced by representative time-series baselines, including LSTM, TCN, Transformer, and DLinear. In this way, the comparison focuses on the quality of the predictive model itself rather than differences in the downstream optimizer.

Table 9 reports the comparative results. The proposed GraphWorldModel_MPC achieved the lowest average 24-step NRMSE of 4.28% and the highest average 24-step R^2 of 0.970 among all compared predictors. More importantly, the predictive advantage was consistently translated into downstream operational gains. Under the same MPC layer, the proposed method reduced the monthly total cost to 5,757,859 RMB, the penalty cost to 2479 RMB, and the grid import to 3060.29 MWh, while also yielding the lowest equivalent carbon emissions and zero reported dispatch violations in the evaluated month. By contrast, the sequence-only and general-purpose time-series baselines exhibited higher multi-step prediction errors, which led to increased cost, larger penalty terms, and weaker closed-loop feasibility.

Table 9. A comparison with advanced time-series prediction baselines under the same MPC interface.

Method	Avg. 24-Step NRMSE (%)	Avg. 24-Step R^2	Total Cost (RMB)	Penalty Cost (RMB)	Grid Import (MWh)	Equivalent Carbon Emissions (tCO ₂)	Dispatch Violation Rate
DLinear-MPC	5.87	0.949	6,017,826	5123	3148.37	3908.26	0.34%
LSTM-MPC	5.34	0.956	5,986,432	4688	3136.55	3894.82	0.27%
TCN-MPC	5.09	0.959	5,941,715	4227	3120.48	3881.36	0.21%
Transformer-MPC	4.84	0.963	5,894,087	3784	3106.92	3864.17	0.14%
GraphWorldModel_MPC	4.28	0.970	5,757,859	2479	3060.29	3781.49	0.00%

These results indicate that the gain of the proposed framework does not come merely from using a nonlinear sequence predictor. Instead, it arises from the joint use of topology-aware spatial encoding, temporal rollout learning, and physics-consistency regularization. Such a design is particularly beneficial in the industrial-park IEHES, where the state evolu-

tion is jointly shaped by structured electricity–heat–hydrogen coupling and intertemporal storage dynamics.

Nevertheless, the results should be interpreted within the tested operating conditions. The performance gain of GraphWorldModel_MPC is most evident when the system exhibits strong cross-carrier coupling, storage carry-over effects, and non-negligible forecast deviations. Under simpler operating conditions with weak coupling or highly accurate forecasts, the advantage of graph-based rollout learning and quantile tightening may become less pronounced. In addition, the improved feasibility is obtained by narrowing selected state and power bounds, which may introduce moderate conservatism when prediction errors are overestimated. Therefore, the proposed framework is most suitable for IEHES dispatch scenarios where predictive accuracy, constraint feasibility, and uncertainty handling must be considered simultaneously.

6. Limitations

Although the proposed GraphWorldModel_MPC demonstrates favorable predictive accuracy, economic performance, and operational feasibility in the tested industrial-park IEHES, several limitations should be acknowledged. First, the performance of the learned world model still depends on the quality, coverage, and representativeness of the training data. If the operating condition changes substantially, such as under abnormal equipment outages, extreme renewable fluctuations, or unseen demand patterns, the learned transition model may experience distribution-shift degradation. Therefore, cross-site validation, online adaptation, and uncertainty calibration under rare operating conditions should be further investigated.

Second, the quantile-based tightening strategy improves closed-loop feasibility by reserving safety margins for selected critical state and power variables, but it may also introduce conservatism when the empirical prediction-error distribution is overestimated or no longer representative of online conditions. In addition, the present study focuses on day-ahead and rolling operational scheduling under fixed device models and parameter settings. Equipment degradation, communication delay, actuator response, and cyber-physical implementation constraints are not explicitly modeled. Future work will therefore focus on adaptive safety-margin updating, deployment-oriented validation, and integration with practical energy management systems.

7. Conclusions

This paper proposed a graph spatiotemporal world-model-driven rolling MPC framework, termed GraphWorldModel_MPC, for low-carbon economic dispatch of industrial-park integrated electricity–heat–hydrogen energy systems. The proposed method addresses multi-energy coupling through directed graph representation and topology-aware message passing, captures intertemporal storage dynamics through temporal encoding and multi-step rollout learning, and mitigates forecast uncertainty through quantile-based safety tightening within a hard-constrained economic MPC. The case studies show that the proposed method achieves high multi-step prediction accuracy and obtains the lowest total operating cost, grid import, equivalent carbon emissions, and penalty cost compared with Conventional EMPC, DRAMPC, and GRU-MPC. The representative-day results further confirm that the obtained dispatch solution maintains electricity-, heat-, and hydrogen-side balance while coordinating battery, thermal storage, and hydrogen storage across different timescales.

The proposed framework is most applicable to industrial-park IEHES scenarios with strong cross-carrier coupling, storage carry-over effects, and non-negligible forecast uncertainty. Its graph representation is extensible to larger multi-energy systems by

adding device nodes and coupling edges, but larger-scale deployment may increase graph-construction complexity, rollout computation, and MPC solving burden. In addition, the learned world model depends on representative historical data, and the quantile-tightening strategy may become conservative if the empirical prediction-error distribution is over-estimated or shifts over time. Future work will focus on cross-site validation, adaptive updating of the learned model and safety margins, and deployment-oriented integration with practical energy management systems.

Author Contributions: J.L.: Methodology; formal analysis; writing—original draft. X.W.: Data curation; validation; supervision. L.W.: Methodology; data curation; formal analysis. Y.S.: Formal analysis; validation. All authors have read and agreed to the published version of the manuscript.

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Conflicts of Interest: Author J.L. and X.W. was employed by the company Inner Mongolia Electric Power (Group) Co., Ltd., Inner, China. Author L.W. was employed by the company Beijing China-Power Information Technology Co., Ltd., Beijing, China. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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